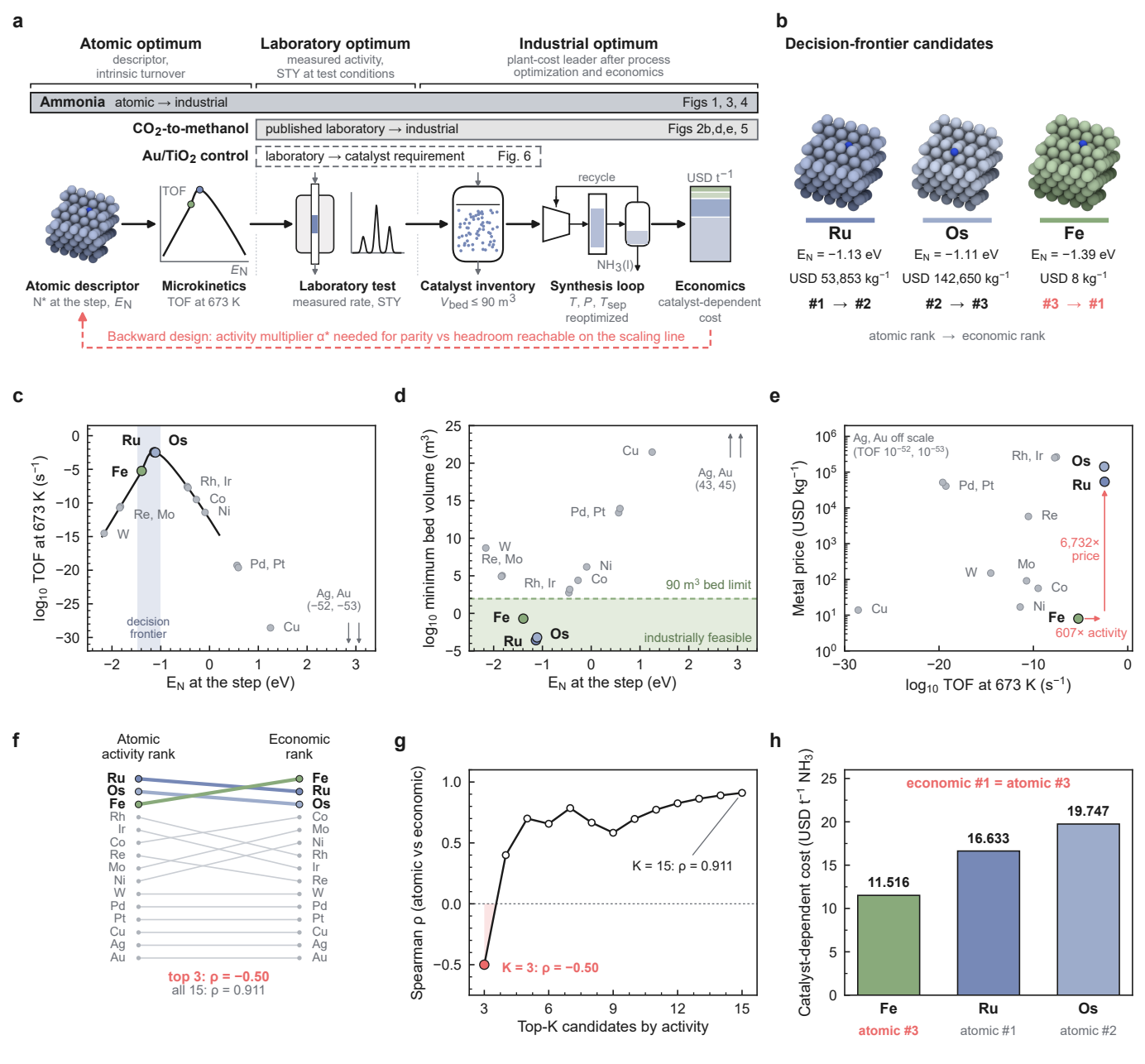

a
Atomic optimum
descriptor, intrinsic turnover
Laboratory optimum
measured activity, STY at test conditions
Industrial optimum
plant-cost leader after process optimization and economics
Ammonia atomic → industrial
Figs 1, 3, 4
CO2-to-methanol
published laboratory → industrial
Figs 2b,d,e, 5
Au/TiO2 control
laboratory → catalyst requirement
Fig. 6
TOF
EN
recycle
NH3(l)
USD t−1
Atomic descriptor
N* at the step, EN
Microkinetics
TOF at 673 K
Laboratory test
measured rate, STY
Catalyst inventory
Vbed ≤ 90 m3
Synthesis loop
T, P, Tsep reoptimized
Economics
catalyst-dependent cost
Backward design: activity multiplier α* needed for parity vs headroom reachable on the scaling line
b
Decision-frontier candidates
Ru
EN = −1.13 eV
USD 53,853 kg−1
#1 → #2
Os
EN = −1.11 eV
USD 142,650 kg−1
#2 → #3
Fe
EN = −1.39 eV
USD 8 kg−1
#3 → #1
atomic rank → economic rank
c
log10 TOF at 673 K (s−1)
Ru
Os
Fe
Rh, Ir
Co
Ni
Re, Mo
W
Pd, Pt
Ag, Au (−52, −53)
Cu
decision frontier
EN at the step (eV)
d
log10 minimum bed volume (m3)
Cu
Ag, Au (43, 45)
Pd, Pt
W
Re, Mo
Ni
Rh, Ir
Co
90 m3 bed limit
Fe
Ru
Os
industrially feasible
EN at the step (eV)
e
Ag, Au off scale (TOF 10−52, 10−53)
Rh, Ir
Os
Ru
Pd, Pt
Re
6,732× price
W
Mo
Co
Cu
Ni
Fe
607× activity
Metal price (USD kg−1)
log10 TOF at 673 K (s−1)
f
Atomic activity rank
Economic rank
Ru Os Fe Rh Ir Co Re Mo Ni W Pd Pt Cu Ag Au
Fe Ru Os Co Mo Ni Rh Ir Re W Pd Pt Cu Ag Au
top 3: ρ = −0.50
all 15: ρ = 0.911
g
Spearman ρ (atomic vs economic)
K = 15: ρ = 0.911
K = 3: ρ = −0.50
Top-K candidates by activity
h
economic #1 = atomic #3
19.747
16.633
11.516
Catalyst-dependent cost (USD t−1 NH3)
Fe
atomic #3
Ru
atomic #1
Os
atomic #2